BENEFITS INTELLIGENCE AGENTS

Benefits Review Next Iteration Requirements Report

Customer requirements, current implementation status, acceptance criteria, and delivery plan for Singapore factual extraction, model experimentation, and execution performance.

Scope: Singapore Company and Market profiles
Catalog: GLB Benefit Datapoints Full V1 (Standard), 160 datapoints
Current benchmark: GPT-5.6 Luna, low reasoning
Prepared: 13 November 2026

1. Executive Summary

> The next iteration has three priorities: improve factual extraction against the finalized Singapore golden dataset, compare GPT-5.6 Luna with GPT-5.6 Sol and GPT-5.5 under controlled conditions, and expose a complete execution profile. Factual Company and Market accuracy remains the acceptance gate before further alignment optimization.

The existing implementation already enforces a maximum of 15 datapoints per call, repeats complete topic evidence in every batch, retries omitted keys, and returned all 160 authoritative GLB V1 records. The latest customer review identified semantic incompleteness inside returned rows rather than missing keys.

Release distinction: prompt `2026-11-extraction-accuracy-v3` and its regression tests are implemented locally but are not yet deployed. The latest customer-reviewed Luna result came from the previous deployed prompt.

2. Requirements Register

Requirement	Current status	Required next action
Maximum 15 datapoints per LLM call	DEPLOYED	Retain unchanged across all benchmark runs.
All 160 authoritative GLB V1 records returned	VALIDATED	Continue structural regression checks.
Complete Company and Market topic context in every batch	DEPLOYED	Confirm the four findings against exact source evidence.
Additional comments and narrative evidence considered	LOCAL V3	Deploy v3 and rerun Luna.
Conditional statistical denominators preserved	LOCAL V3	Verify Healthcare 95.8% wording.
All material measures and shared limits retained	LOCAL V3	Verify Business Travel and Maternity rows.
Semantic completeness audit and retry	NOT IMPLEMENTED	Add only if the v3 Luna baseline remains incomplete.
Controlled Luna, Sol, and GPT-5.5 comparison	NOT STARTED	Run identical, repeated benchmark executions.
Per-batch and pipeline-phase telemetry	NOT IMPLEMENTED	Instrument before the model benchmark.
Progressive availability of usable results	NOT IMPLEMENTED	Measure current behavior, then decide whether row/batch streaming is required.

3. Workstream 1: Factual Extraction Accuracy

Customer findings

Area	Observed issue	Acceptance criterion
EAP	Childcare assistance, medical advice, and training returned `Not Provided` despite Wellbeing evidence.	Each affected row contains the correct source-supported detail and is assigned to the correct datapoint.
Healthcare	Hospitalization maximum statistic was presented as applying to all peers.	The result states that 95.8% applies to organizations with a maximum.
Business Travel	Summary omitted the Company S$10M out-of-country medical maximum and Market S$300K median accidental-death benefit.	Both measures appear with the correct party, benefit, unit, and basis.
Maternity	Individual service summaries omitted the shared Company S$5,000 annual maximum.	Every affected individual-service row includes the shared annual maximum without applying it outside its source scope.

Diagnosis

- Topic slicing is unlikely to be the primary cause because each batch receives the complete relevant Company and Market topic sections.
- Structured parsing copies returned Company and Market strings and did not report dropped keys.
- Existing retry handles absent keys only; it does not detect a present row with an incomplete value.
- The strongest identified cause is model interpretation combined with earlier prompt competition: additional comments were de-prioritized, F4 encouraged compression, and shared or multiple measures were not explicit enough.

Implementation and verification

1. Deploy prompt v3 with rules for additional comments, exact conditional denominators, multiple measures, shared limits, and concise-but-complete F4 output.
2. Rerun GPT-5.6 Luna at low reasoning using the identical source files, catalog, batches, and concurrency.
3. Compare all 160 rows against the finalized golden dataset by fact and source support, not text similarity.
4. Classify the four known examples as Passed, Partially passed, or Failed, recording source evidence, golden value, and actual value.
5. Repeat the stable configuration at least three times to assess consistency.

Conditional semantic audit

If v3 remains incomplete, introduce a targeted verification pass for complex or suspicious rows. The audit must receive the original datapoint prompt, full topic evidence, and initial value; return changes only when explicit evidence supports them; preserve the initial value otherwise; and report audited rows, recovered facts, tokens, and latency separately. It must not blindly retry every `Not Provided` value.

4. Workstream 2: Controlled Model Experimentation

The customer requested a direct comparison of the current Luna baseline with Sol and potentially GPT-5.5. These deployments are currently available and in a succeeded state.

Candidate	Initial configuration	Role
GPT-5.6 Luna	Low reasoning	Current baseline
GPT-5.6 Sol	Same supported reasoning configuration	Primary comparison
GPT-5.5	Same supported reasoning configuration	Secondary comparison

Every model run must use the same Company profile, Market profile, GLB V1 catalog and order, prompt version, 15 batch definitions, concurrency limit, code/container version, and validation rules. Reasoning effort is a separate experimental variable and must not be changed inside a model-only comparison.

Measures

- Factual accuracy and prompt-level completeness against the golden dataset
- Consistency across at least three repeated runs
- Wrong-benefit attribution, denominator errors, and unsupported facts
- Total wall-clock time, per-batch model time, queue time, and retry time
- Input, output, reasoning, and total tokens
- Estimated cost using the applicable deployment pricing

The shared deployment inventory represents models currently deployed in the environment. Other models may be deployable subject to Azure regional availability, quota, access, and WTW approval.

5. Workstream 3: Execution Performance and Parallelism

Current behavior

- All 15 batch tasks are scheduled together.
- An application semaphore permits at most nine active batch tasks, so 15 batches require at least two scheduling waves.
- An omitted-key retry occurs inside the same bounded task and continues to occupy its concurrency slot.
- `processing_time_ms` is application wall-clock time from before slicing through final aggregation.
- Current per-topic timing is the sum of batch durations for that topic. Concurrent durations overlap and must not be added to derive wall-clock time.
- The SSE endpoint sends an immediate status event, but usable rows and topic-complete events are emitted only after the full pipeline returns.

> **Interpretation of 36+ seconds:** this is the current end-to-end application duration, not the sum of individual batch times. The customer's theoretical 12-second lower bound assumes all 15 calls execute concurrently without Azure-side queuing; the current application cap is nine.

Required telemetry

Level	Required measurements
Batch	Topic, batch index/count, queued/start/end timestamps, attempt number, retry reason, model duration, parse duration, validation duration, and outcome.
Tokens	Input, output, reasoning where exposed, and total tokens for every attempt.
Concurrency	Active request count at start/end, observed peak concurrency, semaphore wait time, throttling, and retry-after information.
Pipeline	Slicing, LLM wall time, parsing, factual validation, deterministic alignment, aggregation, serialization, and total duration.
Availability	Time to first status, first completed batch, first usable result, and complete 160-row result.

Acceptance criteria

- Batch timestamps demonstrate actual overlap and measured peak concurrency.
- Phase durations reconcile with total wall-clock time within a documented tolerance.
- Initial calls and retries are independently identifiable.
- Token totals reconcile with the invocation-level total.
- Azure throttling or queuing is reported as evidence, not inferred from latency alone.
- Model comparison reports use the same telemetry schema.

6. Delivery Sequence

Phase	Activities	Exit condition
1. V3 baseline	Build, deploy, invoke Luna, and verify metadata and all 160 rows.	Four known findings scored against source and golden evidence.
2. Instrumentation	Add batch, token, concurrency, phase, retry, and availability telemetry.	One Luna run produces a reconciled execution profile.
3. Semantic audit decision	Review v3 failures and implement targeted audit only if needed.	No unresolved known issue without a recorded reason or review flag.
4. Model benchmark	Run Luna, Sol, and GPT-5.5 at least three times under controlled conditions.	Comparable quality, consistency, performance, token, and cost matrix.
5. Recommendation	Select the preferred model/configuration and document trade-offs.	Customer-ready recommendation supported by reproducible evidence.

7. Required Inputs and Deliverables

Inputs

- Singapore Company profile and Market profile used for the customer-reviewed run
- Finalized Singapore golden dataset
- Authoritative GLB V1 runtime catalog
- Latest customer-reviewed result JSON and new v3 result JSON
- Applicable Azure model pricing and any quota or throughput constraints

Deliverables

- Four-finding factual verification record and complete 160-row scoring output
- Detailed execution profile with batch and phase telemetry
- Controlled Luna/Sol/GPT-5.5 benchmark matrix
- Recommended production model and reasoning configuration
- Known limitations, residual review items, and deployment traceability

8. Definition of Done

The iteration is complete when the selected configuration returns all 160 authoritative rows; satisfies the agreed factual accuracy and completeness threshold; resolves or explicitly flags the four known findings; produces reproducible timing, token, retry, and concurrency evidence; and has been compared fairly across the selected models. Alignment optimization should resume only after this factual gate is accepted.

This document defines the next iteration requirements and does not claim that locally implemented prompt v3 changes, semantic auditing, detailed telemetry, or controlled model benchmarks are already deployed or complete.